

Proof. We'll use the real JCF here (more details: <https://www.researchgate.net/publication/360359411>)
 Suppose $A = P^{-1}JP$, where $J = \text{diag}(D_1, D_2, \dots, D_K)$, and

$$D_i = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \dots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

$$\text{or} \begin{pmatrix} |\lambda| \cos \theta & |\lambda| \sin \theta & 1 & 0 & & & \\ -|\lambda| \sin \theta & |\lambda| \cos \theta & 0 & 1 & & & \\ & |\lambda| \cos \theta & |\lambda| \sin \theta & 1 & 0 & & \\ & -|\lambda| \sin \theta & |\lambda| \cos \theta & 0 & 1 & & \\ & & \dots & & & |\lambda| \cos \theta & |\lambda| \sin \theta & 1 & 0 \\ & & & & & -|\lambda| \sin \theta & |\lambda| \cos \theta & 0 & 1 \\ & & & & & & |\lambda| \cos \theta & |\lambda| \sin \theta \\ & & & & & & -|\lambda| \sin \theta & |\lambda| \cos \theta \end{pmatrix}.$$

Suppose $D_i = D'_i I'$, and $I' = \begin{pmatrix} & & 1 \\ & \dots & \\ 1 & & \end{pmatrix}$, then $J = J' I'$. Notice that D'_i and I' is symmetric, so $A = P^{-1}JP = (P^{-1}J'(P^{-1})^T)(P^T I' P)$ is the product of two real symmetric matrix. $\square$